AI-Powered Academic Project Manager

College students often manage multiple courses with overlapping deadlines, group projects, exams, and shifting priorities. Traditional tools, such as to-do lists and calendars, require constant manual updates and don't automatically adapt to deadline changes, workload increases, or unexpected events. As a result, students experience missed deadlines, inefficient time use, and unnecessary stress. This problem is especially relevant for those enrolled in project-based or more rigorous classes, where assignments often span multiple weeks and involve task dependencies. Inefficient academic planning leads to lower academic performance, increased stress, and poor time management outcomes, making this a productivity and performance problem rather than merely a personal inconvenience.

The primary users are undergraduate college students managing multiple concurrent assignments across different courses. Secondary users may include graduate students or working students who balance academic responsibilities with jobs or internships.

An agentic AI system is appropriate because academic planning is not a one-time task but an ongoing process that requires monitoring, decision-making, and adaptation. Unlike a simple chatbot or static planner, an agentic system can break down large projects into actionable tasks, monitor progress approaching deadlines, re-plan the schedule when constraints change, and critically review completed work. The autonomy and multi-step reasoning make an agentic AI solution more effective than a traditional human-driven or rule-based workflow.

User case and scenarios

Scenario 1: Planning a multi-week assignment

A student uploads an assignment description and rubric for a final project. The system not only generates an initial task breakdown and timeline, but also stores the plan as an active project state. As the student marks tasks complete or misses planned milestones, the system continuously monitors progress, updates task status, and proactively alerts the student when they fall behind or when task dependencies are at risk. Rather than requiring repeated user prompts, the system autonomously evaluates whether the current plan remains feasible and initiates adjustments when necessary. This system maintains a persistent project state, provides ongoing monitoring, and issues proactive alerts without user prompting.

Scenario 2: Adapting to Schedule Changes

Midway through the semester, a student receives a new exam date and an additional assignment. The system automatically detects conflicts between existing project timelines and the new constraints. Without explicit user instruction, the system evaluates trade-offs, reprioritizes tasks, and generates a revised schedule that minimizes deadline risk. The system then presents the updated plan, along with a justification for the changes, and continues monitoring the revised schedule for feasibility. The system detects changes autonomously, replans without being told how, explains its decisions, and continues monitoring after replanning.

Agents and Roles

The initial system design would include the planner agent, monitoring agent, re-planning agent, and critic agent. The planner agent would break down the assignments into tasks, estimate effort, and create an initial timeline. The monitoring agent would track deadlines, task completion, and progress updates. Dynamically adjusting schedules when new constraints or conflicts arise is the job of the replanning agent. Finally, the critic agent will review final submissions against the rubrics and provide feedback.

Tools, APIs, and Data Sources

Tools, APIs, and Data Sources that would be used include calendar integration or a simulated calendar. Local file uploads for syllabi, rules, and assignments. Task management interface(web-based front end). Optional document-parsing tools for PDFs or text files are another option.

Some potential challenges include hallucinations when interpreting unclear assignment instructions, inaccurate time estimates for tasks, long context handling for large syllabi or rubrics, and tool failures or API rate limits. While the constraints include privacy concerns when handling academic documents, a limited project timeline within a single semester, and computational limits for multi-agent execution. As for the Mitigation strategies, it would require user confirmation for generated plans, use structured prompts and role-specific agents, limit scope to a small number of classes or assignments, and use local files instead of external API's when possible.

The system's expected outcomes will be evaluated based on task-planning accuracy, user satisfaction, perceived usefulness, reduction in missed or rushed deadlines, and a comparison of planned and actual task-completion timelines. This evaluation will be expanded in later milestones using pilot testing and user feedback.